

ABC-Machine Learning For Prediction

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Model1: Logistic Regression

Evaluation metrics: LR-Accuracy

Classification Report:

	precision	recall	f1-score	support
0	0.91	0.87	0.89	270
1	0.82	0.88	0.84	177

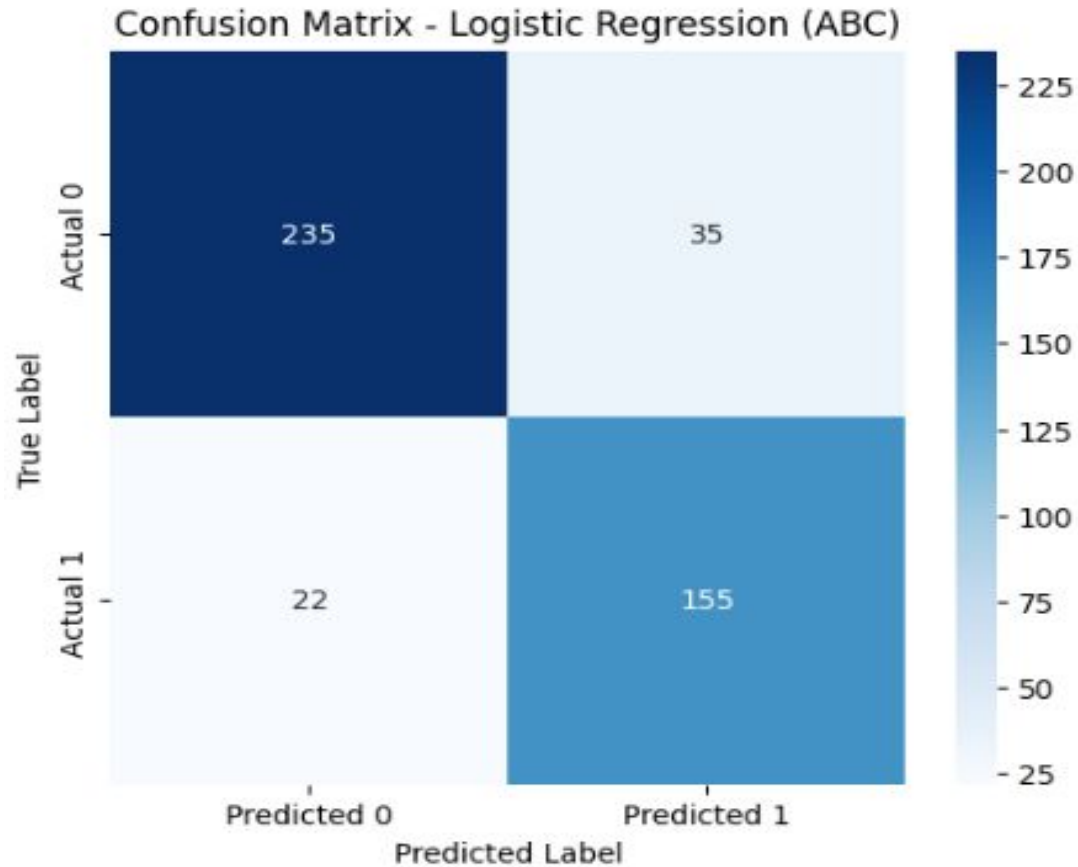
accuracy			0.87	447
macro avg	0.87	0.87	0.87	447
weighted avg	0.88	0.87	0.87	447

Evaluation metrics: LR-Accuracy

Classification Report:

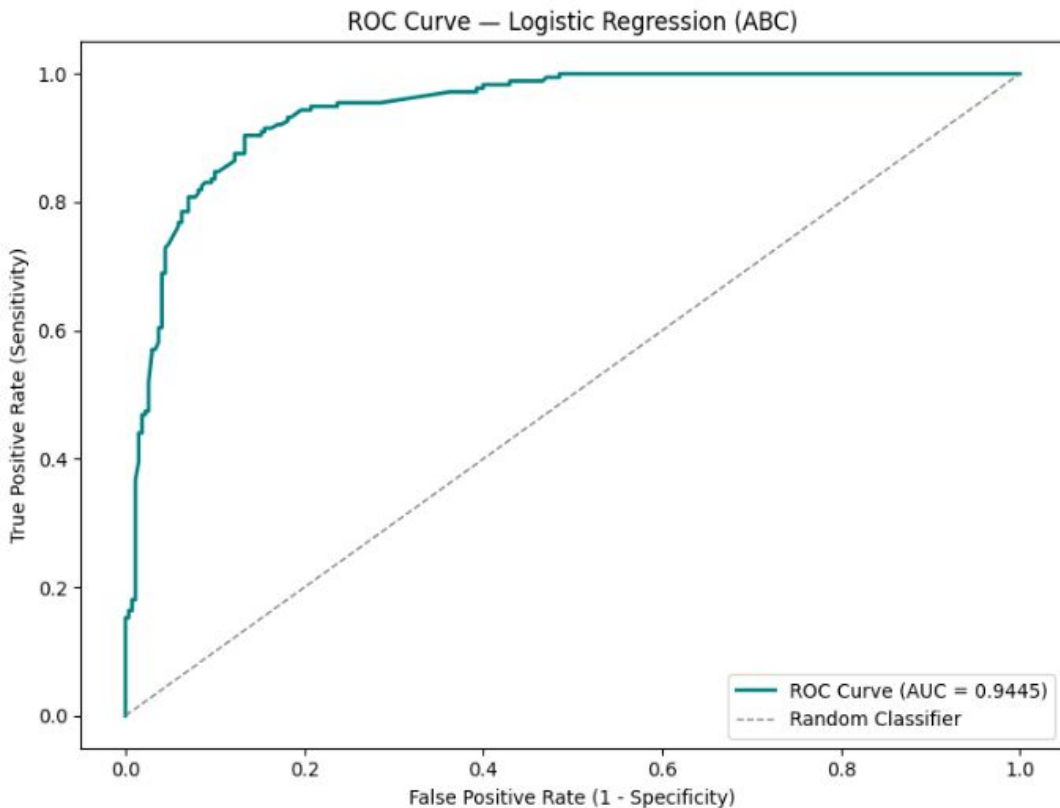
	precision	recall	f1-score	support
0	0.91	0.87	0.89	270
1	0.82	0.88	0.84	177
accuracy			0.87	447
macro avg	0.87	0.87	0.87	447
weighted avg	0.88	0.87	0.87	447

Confusion Matrix - Logistic Regression



Precision, Recall, F1 score, precision- Recall & **ROC AUC curves**

- Data set is balanced - (our dataset:)
- False positives and False negatives are of equal importance #Critical for our model



Precision- Recall & ROC AUC curves

Model 2: Random Forest Classifier

Evaluation metrics: RF-Accuracy

--- Classification Report ---

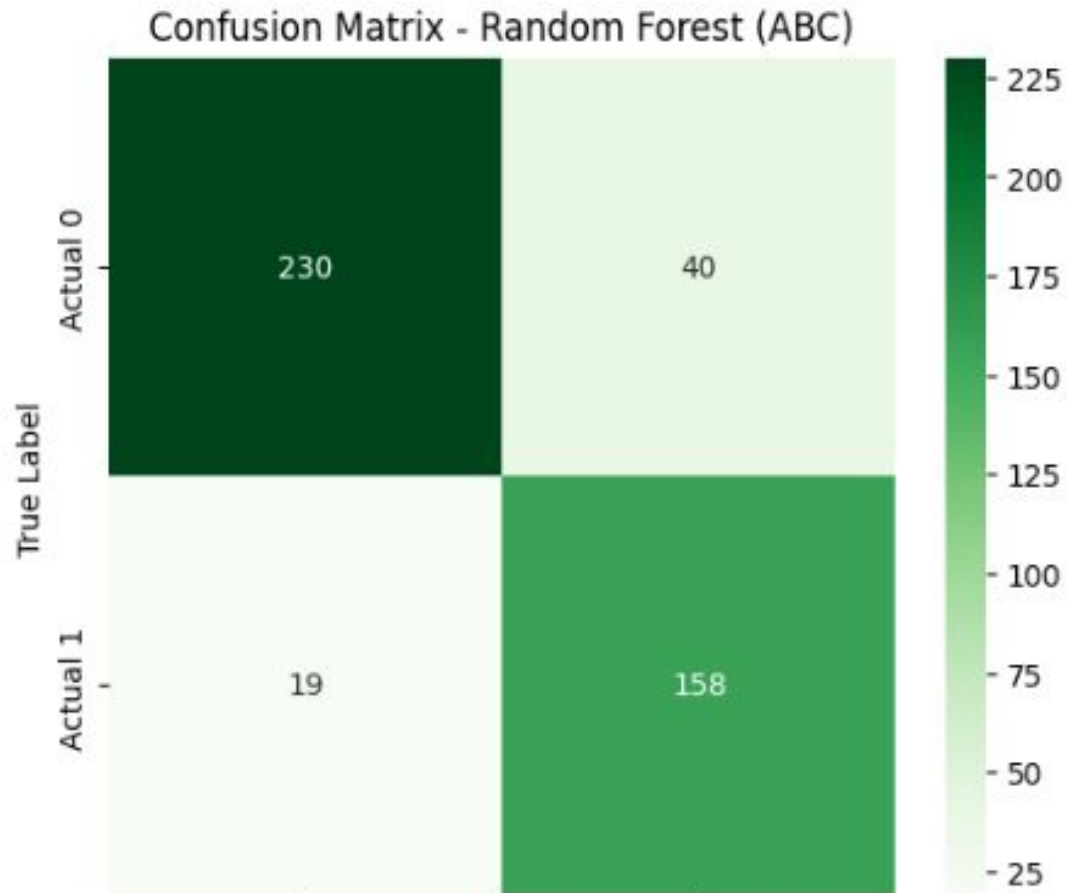
	precision	recall	f1-score	support
0	0.92	0.85	0.89	270
1	0.80	0.89	0.84	177
accuracy			0.87	447
macro avg	0.86	0.87	0.86	447
weighted avg	0.87	0.87	0.87	447

Evaluation metrics: RF-Accuracy

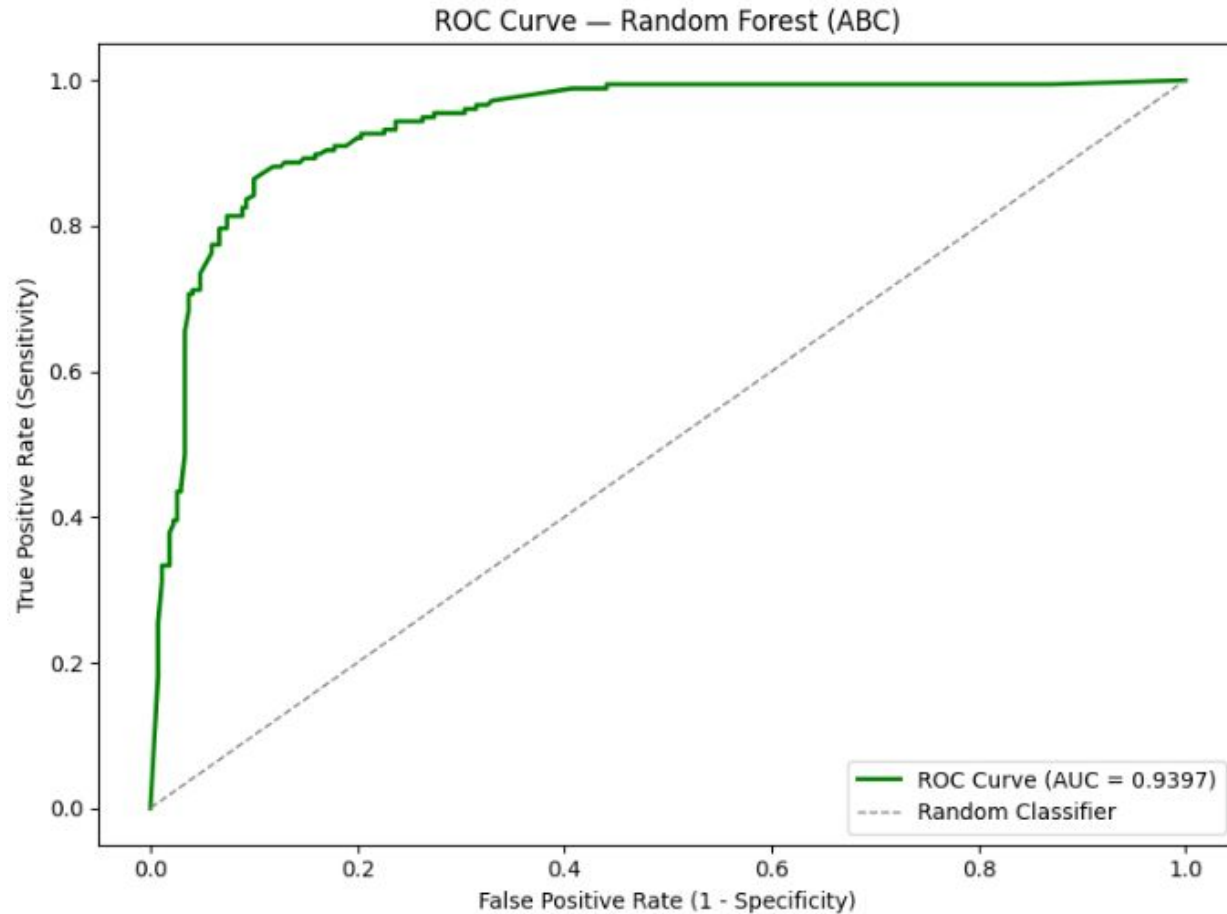
--- Classification Report ---

	precision	recall	f1-score	support
0	0.92	0.85	0.89	270
1	0.80	0.89	0.84	177
accuracy			0.87	447
macro avg	0.86	0.87	0.86	447
weighted avg	0.87	0.87	0.87	447

Confusion Matrix - Random Forest



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves

Model 3: Support Vector Machine

Evaluation metrics: SVM-Accuracy

--- SVM Test Classification Report ---

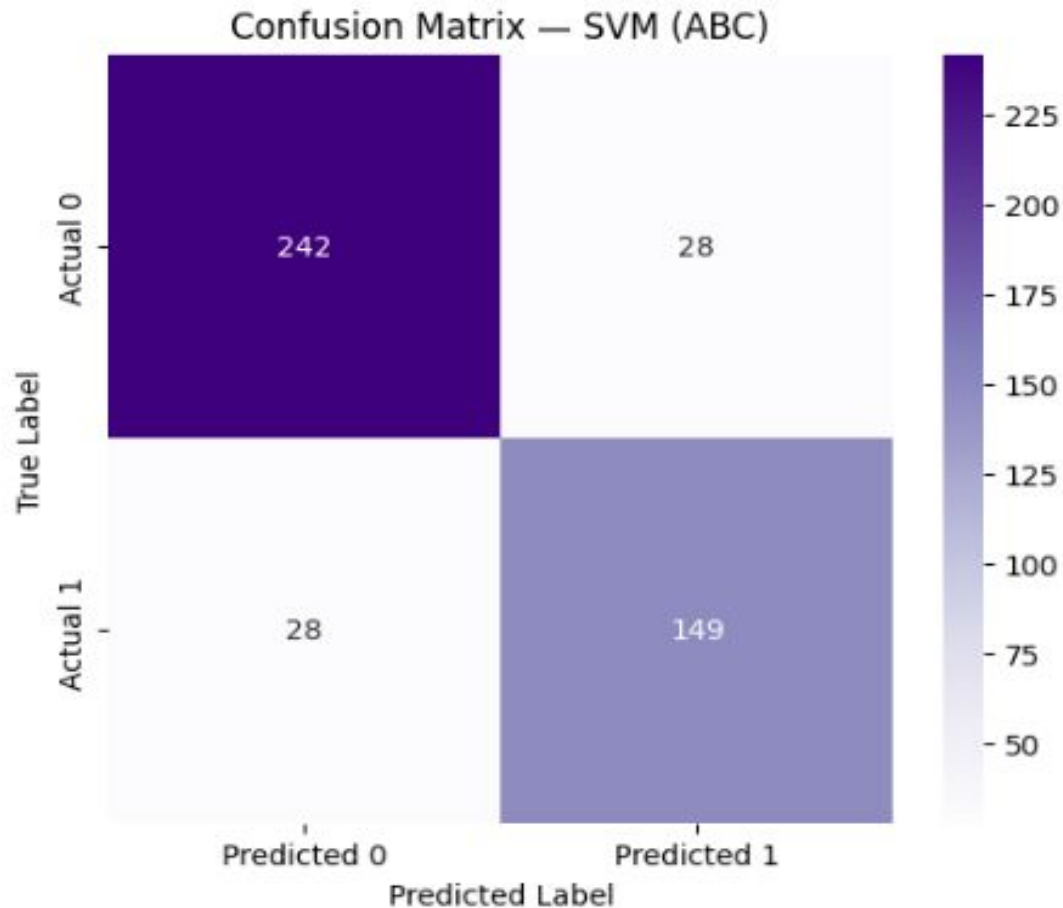
	precision	recall	f1-score	support
0	0.90	0.90	0.90	270
1	0.84	0.84	0.84	177
accuracy			0.87	447
macro avg	0.87	0.87	0.87	447
weighted avg	0.87	0.87	0.87	447

Evaluation metrics: SVM-Accuracy

--- SVM Test Classification Report ---

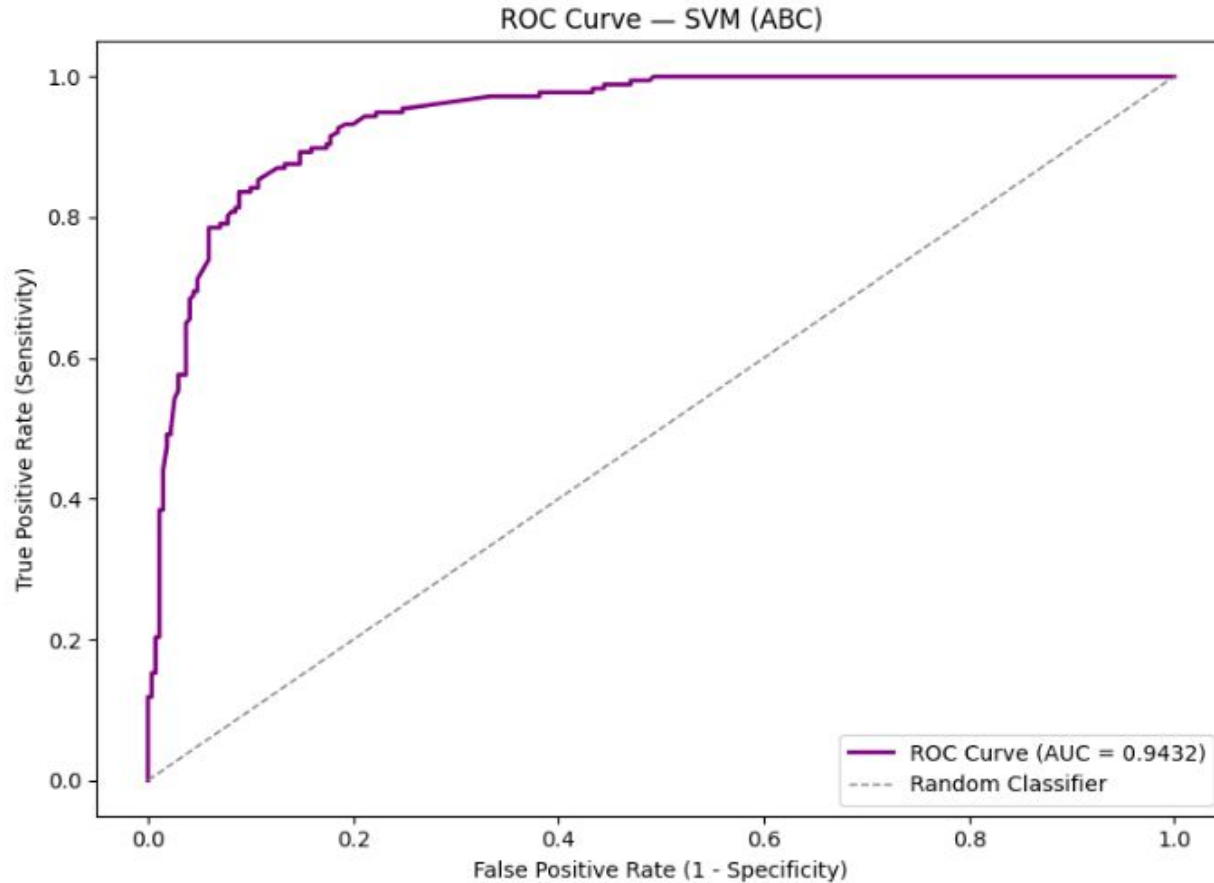
	precision	recall	f1-score	support
0	0.90	0.90	0.90	270
1	0.84	0.84	0.84	177
accuracy			0.87	447
macro avg	0.87	0.87	0.87	447
weighted avg	0.87	0.87	0.87	447

Confusion Matrix - Support Vector Machine



Precision- Recall & ROC AUC curves

Precision- Recall & ROC AUC curves



Model 4: Extreme Gradient Boost

Evaluation metrics: XGB-Accuracy

--- XGBoost Test Classification Report ---

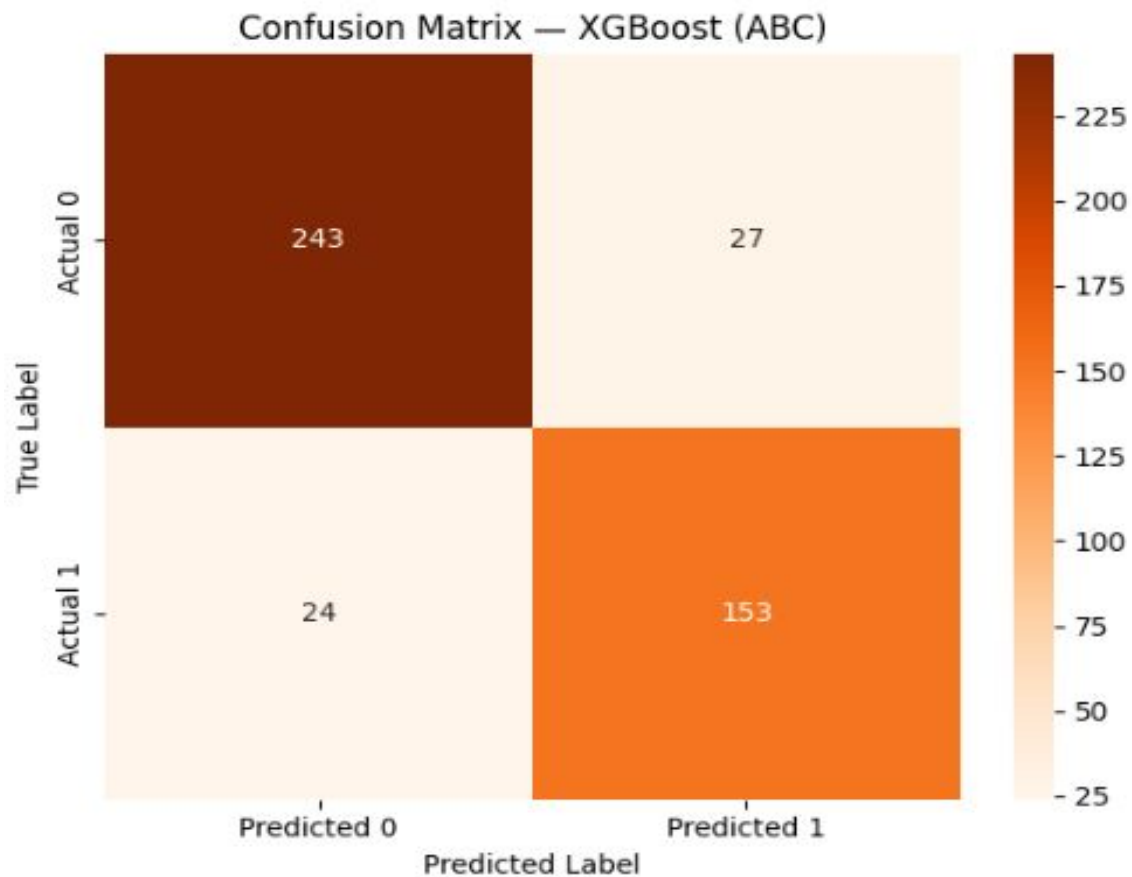
	precision	recall	f1-score	support
0	0.91	0.90	0.91	270
1	0.85	0.86	0.86	177
accuracy			0.89	447
macro avg	0.88	0.88	0.88	447
weighted avg	0.89	0.89	0.89	447

Evaluation metrics: XGB-Accuracy

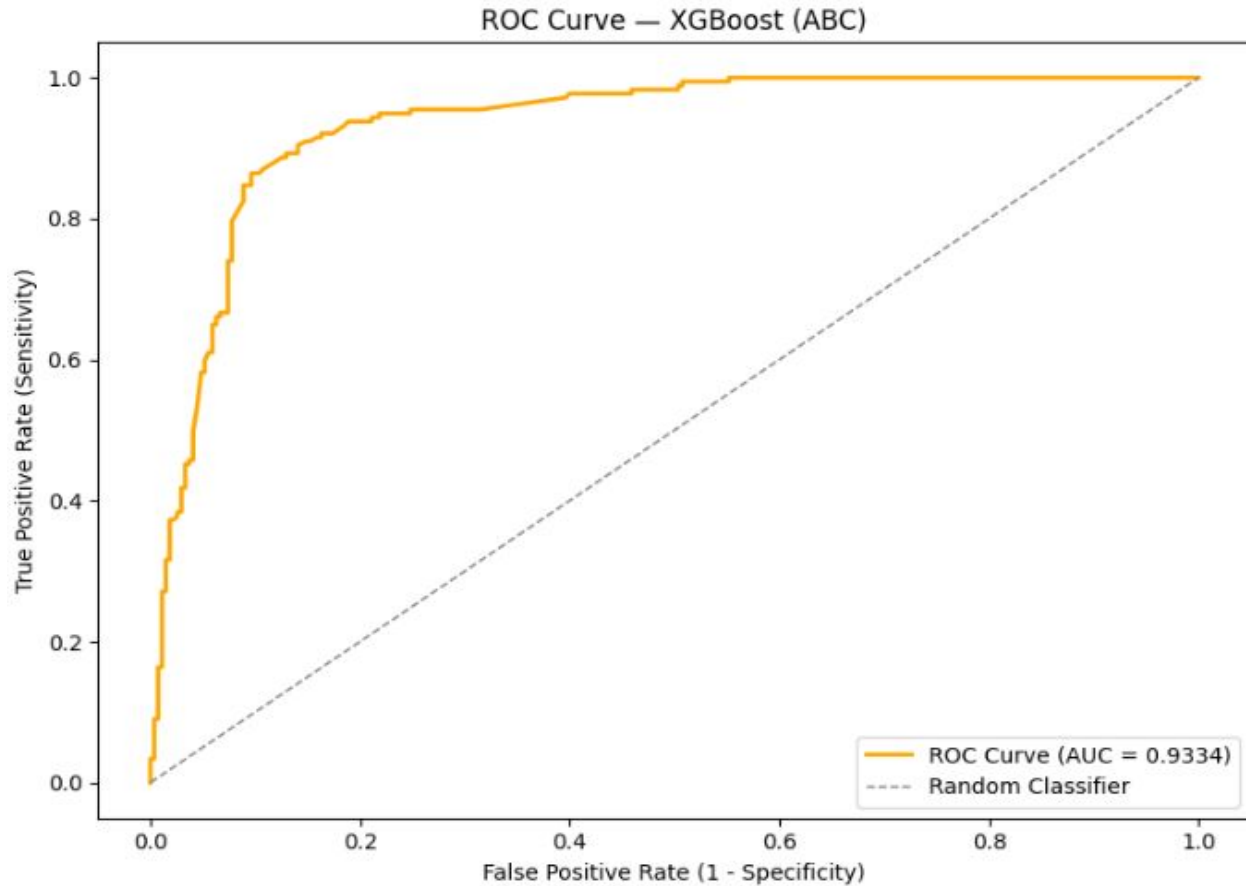
--- XGBoost Test Classification Report ---

	precision	recall	f1-score	support
0	0.91	0.90	0.91	270
1	0.85	0.86	0.86	177
accuracy			0.89	447
macro avg	0.88	0.88	0.88	447
weighted avg	0.89	0.89	0.89	447

Confusion Matrix - XGBOOST



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves

Comparison (Macro -avg) - ABC

Model	Precision	Recall	F1 score
LR	0.87	0.87	0.87
RF	0.86	0.87	0.87
SVM	0.87	0.87	0.87
XGB	0.88	0.88	0.88

Model Ranking by MCC (ABC)

Model Ranking by MCC (ABC)

Rank	Model	MCC
1	XGB	0.7623
2	SVM	0.7381
3	LR	0.7381
4	RF	0.7330

Best model : XGB